

Multiple Choice (10 marks)

1. The sub key length at each round of DES is_____
 - (a) 32
 - (b) 56
 - (c) 48
 - (d) 64
2. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Dark web
 - (d) Surface web
3. Which of the following styles of fuzzer is more likely to explore paths covering every line of code in the following program?
 - (a) Generational
 - (b) Blackbox
 - (c) Whitebox
 - (d) Mutation-based
4. A digital signature needs a
 - (a) Private-key system
 - (b) Shared-key system
 - (c) Public-key system
 - (d) All of them
5. A/an _____ is a program that steals your logins & passwords for instant messaging applications.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) Trojan-Downloader
 - (d) Ransom Trojan

True / False (10 marks)

Answer True or False

6. True or False: Maltego is primarily designed for wireless traffic sniffing and does not focus on link analysis.
7. True or False: The message must be encrypted and decrypted at the sender site for secure communication to occur.
8. True or False: Penetration testing is a comprehensive evaluation of a system for security flaws and vulnerabilities.
9. True or False: A hash function guarantees that a message cannot be replaced by an attacker without detection.
10. True or False: The Deep Web comprises all online content that cannot be indexed by traditional search engines.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the height of the given binary tree.
12. Write a function to check if given tuple is distinct or not.

End of Paper